

KILSBERGEN-SUTTARBODA, Sweden

WMO: 024520

Lat: 59.2996N

Lon: 14.8956E

Elev: 219

StdP: 98.72

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-11.9	-17.1	0.9	-12.0	-14.7	1.1	-10.2	10.4	4.7	9.2	3.6	2.7	310	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	25.3	17.2	23.6	16.5	21.9	15.7	18.7	22.8	17.7	21.5	16.8	20.4	2.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.8	19.9	16.4	11.9	19.2	15.3	11.1	18.2	54.0	22.6	50.8	21.7	48.0	20.5	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.9	6.1	-16.2	27.7	2.9	1.6	-18.3	28.8	-20.0	29.8	-21.6	30.7	-23.7	31.9		
			-16.6	19.9	2.6	1.1	-18.5	20.7	-20.0	21.4	-21.5	22.0	-23.4	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-3.2	-2.9	0.2	5.5	10.4	14.1	16.8	15.6	11.7	5.9	1.6	-1.0
	DBStd	7.94	4.57	4.17	4.11	3.58	3.86	3.12	3.03	2.64	2.86	3.45	3.79	4.60
	HDD10.0	2002	409	360	303	143	41	3	0	1	16	133	251	342
	HDD18.3	4434	668	593	561	386	246	132	68	92	200	386	501	600
	CDD10.0	644	0	0	0	7	53	127	212	174	66	6	0	0
	CDD18.3	36	0	0	0	0	1	6	22	8	0	0	0	0
	CDH23.3	169	0	0	0	0	6	31	107	25	0	0	0	0
	CDH26.7	14	0	0	0	0	0	2	11	2	0	0	0	0
Wind	WSAvg	3.1	3.4	3.2	3.4	3.2	3.0	2.9	2.7	2.5	2.9	3.1	3.3	3.6
Precipitation	PrecAvg	747	54	43	35	43	64	64	86	77	66	74	69	56
	PrecMax	1037	121	95	64	86	140	122	212	174	137	182	168	114
	PrecMin	578	6	19	3	12	32	25	17	8	16	0	24	9
	PrecStd	102	30	20	16	22	27	31	47	39	36	42	35	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.6	12.9	18.7	24.3	26.5	28.1	25.8	21.2	14.8	10.6	8.7
		MCWB	5.9	4.8	7.5	10.5	15.0	16.5	18.3	17.4	15.6	12.4	9.9	7.5
	2%	DB	5.0	4.9	9.6	16.2	21.7	23.9	26.0	23.6	19.0	12.9	9.2	7.1
		MCWB	4.2	3.6	5.3	8.9	13.2	15.6	18.0	17.4	14.5	11.5	8.3	6.1
	5%	DB	3.8	3.7	7.4	13.7	19.5	21.9	24.4	21.8	17.2	11.8	8.1	5.9
		MCWB	3.3	2.7	4.2	8.1	12.7	14.7	17.3	16.6	13.9	10.7	7.2	4.9
	10%	DB	2.7	2.6	5.6	11.4	17.1	19.9	22.5	20.1	15.8	10.8	6.9	4.8
		MCWB	2.2	1.9	3.3	6.6	11.5	13.7	16.5	15.7	13.1	9.8	6.2	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.6	7.6	11.8	16.6	17.9	20.5	19.8	16.8	13.2	10.2	8.2
		MCDB	6.4	6.1	12.0	17.2	21.5	22.5	24.6	23.1	19.6	14.0	10.5	8.4
	2%	WB	4.6	4.0	5.8	9.8	15.0	16.7	19.1	18.6	15.5	12.0	8.7	6.5
		MCDB	4.9	4.5	8.5	14.6	18.9	22.0	23.6	21.7	18.1	12.6	9.1	6.9
	5%	WB	3.4	3.0	4.8	8.5	13.5	15.6	18.2	17.4	14.5	10.9	7.5	5.2
		MCDB	3.8	3.6	6.8	12.8	17.6	20.4	22.7	20.0	16.8	11.6	8.0	5.6
	10%	WB	2.3	1.9	3.7	7.2	12.2	14.5	17.3	16.5	13.6	9.9	6.3	4.1
		MCDB	2.7	2.5	5.2	10.6	16.2	18.7	21.3	19.2	15.5	10.7	6.8	4.6
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.1	5.8	7.5	7.9	7.7	7.4	6.7	6.0	4.7	3.6	3.8
		MCDBR	4.5	4.6	8.3	10.3	10.9	10.2	9.6	8.6	7.6	5.2	4.8	4.2
		MCWBR	4.6	3.7	5.2	5.6	5.7	5.0	4.3	4.1	4.3	4.0	4.7	4.4
	5% WB	MCDBR	4.4	4.1	7.5	9.4	9.4	9.2	8.5	7.4	7.0	4.8	4.6	4.4
		MCWBR	4.5	3.5	5.1	5.5	5.6	5.0	4.4	4.4	4.5	4.1	4.5	4.7
Clear-Sky Solar Irradiance	taub		0.267	0.307	0.329	0.366	0.362	0.352	0.374	0.366	0.335	0.325	0.291	0.249
	taud		2.223	2.309	2.372	2.339	2.413	2.459	2.415	2.433	2.508	2.446	2.287	2.143
	Ebn at Noon		544	710	811	836	868	882	851	830	803	687	507	417
	Edh at Noon		43	67	85	104	104	101	103	94	75	59	41	31
All-Sky Solar Radiation	RadAvg		0.31	0.92	2.45	3.89	4.98	5.65	5.10	4.02	2.69	1.20	0.40	0.21
	RadStd		0.05	0.17	0.29	0.45	0.47	0.35	0.59	0.44	0.35	0.21	0.09	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-135	-162	N/A	N/A

Nomenclature: See separate page